

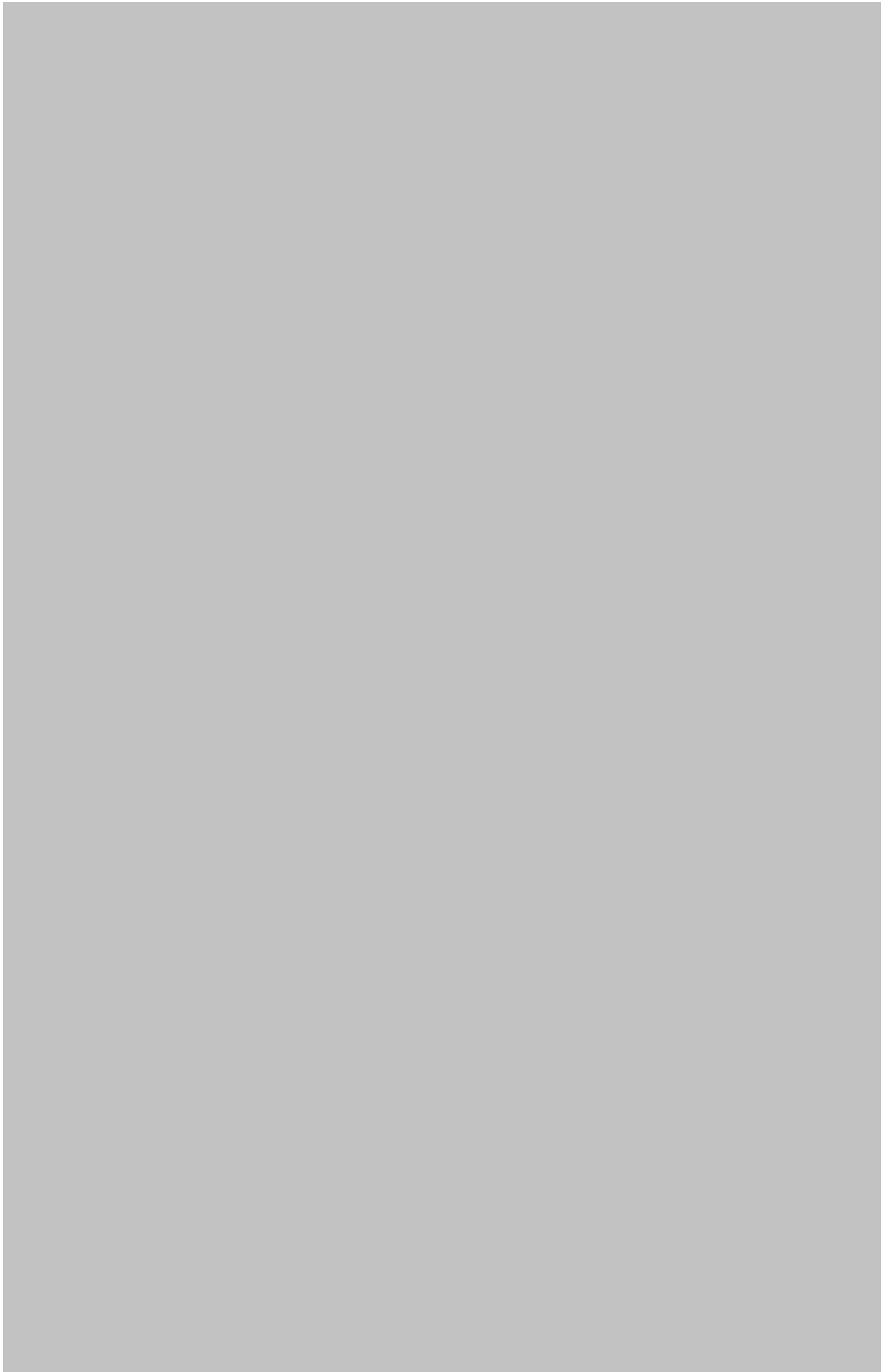


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Chapter 01: Problem Statement

Autism spectrum disorder (ASD) is a complex neurological condition that affects how children behave, interact with others, and communicate. Early detection and intervention are crucial to enhance long-term outcomes and help affected children develop more socially and cognitively. However, autism is often misdiagnosed or only discovered when significant developmental disabilities are evident. This delay is often caused by the current diagnostic approach, which mainly relies on subjective behavioral observations, clinical expertise, and protracted evaluation procedures conducted by experts. Because they are not identified until later, many children miss the crucial early intervention phase, especially in developing countries like Sri Lanka.

Due to ignorance or a lack of access to trustworthy screening tools, parents frequently find it difficult to recognize early signs of autism. Pediatricians and general practitioners might not have the time or resources to conduct in-depth behavioral evaluations. The lack of data-driven diagnostic support tools, which are frequently costly, time-consuming, and unable to scale to large populations, contributes to underdiagnosis and treatment delays. There is a substantial gap between early detection and the diagnostic tools that are currently available.

As artificial intelligence and data science have grown quickly, machine learning techniques have demonstrated promise in spotting intricate developmental and behavioral patterns that could point to autism. Nevertheless, there are few localized models appropriate for Sri Lankan or comparable regional contexts, and the use of such technologies in healthcare is still restricted. A useful, trustworthy, and interpretable machine learning model is required in order to help parents and medical professionals identify early indicators of autism using readily available data, such as behavioral observations, questionnaires, or developmental indicators.

The initiative aims to address the lack of an efficient, cost-effective, and evidence-based approach for early autism screening in children. It aims to develop a machine learning-based model to identify potential autism risks at an early age, improving the accuracy of early screening and promoting timely treatments, ultimately enhancing the developmental outcomes and quality of life for children with autism spectrum disorder.

Chapter 02: Project Description

Background and Rationale

A neurodevelopmental disorder that affects behavior, interaction, and communication, autism spectrum disorder (ASD) is a lifelong condition. One out of every 100 children worldwide are impacted. Results are improved by early detection and intervention. However, a lot of kids, particularly in underdeveloped nations like Sri Lanka, don't receive a diagnosis until they exhibit obvious behavioral and academic issues, which usually happen when they're in school.

Autism is diagnosed through behavioral assessments, parental observations, and clinical judgment by pediatricians or psychologists. However, these methods are labor-intensive and dependent on experts, often leading to delays in diagnosis, especially in rural areas. Parents may lack knowledge or resources.

Advancements in machine learning and artificial intelligence have enabled data-driven early detection in healthcare, enabling early autism screening by analyzing children's behavioral and developmental data. This provides parents and healthcare providers with a reliable, practical, and accessible tool for early autism screening.

The goal of the project is to develop a machine learning-based system for the early diagnosis of autism spectrum disorder in children by utilizing behavioral, social, and developmental markers. Rather than assuming the role of clinical diagnosis, the goal is to guide early action.

Objectives

Main Objective

To develop a machine learning-based model that, using behavioral and developmental data, can effectively and precisely forecast a child's risk of developing autism at a young age.

Specific Objectives

The project aims to improve early autism screening by using publicly accessible datasets or standardized questionnaires. It will find, clean, and normalize features for optimal model performance using machine learning algorithms like Random Forest, Decision Trees, SVM, and Logistic Regression. The goal is to provide an easy-to-use interface for parents and doctors to input behavioral and developmental data,

facilitating early risk assessments. The proposed model will be evaluated and validated to ensure accuracy and reliability in medical settings. The project aims to provide affordable, evidence-based early autism screening options for both urban and rural populations.

Scope

The project aims to develop an early autism detection prototype and machine learning prediction model, analyzing age, gender, behavioral responses, social interactions, and communication abilities.

Included in the scope are:

- Gathering and preprocessing datasets concerning autism.
- Machine learning model development and comparison.
- Putting the top-performing model into practice in a working prototype.
- Accuracy-based performance metrics are used for system testing and evaluation.

Not included in the scope:

- Direct medical diagnosis or the substitution of expert psychological evaluations.
- Gathering genetic or clinical information from actual patients because of privacy and ethical restrictions.
- extensive implementation or integration (only at the prototype level) with hospital systems.

Expected Outcomes

The following results are anticipated at the project's conclusion:

- A prototype system is being developed for physicians and parents to input behavioral data and receive prediction results.
- Thorough documentation that covers performance analysis, model evaluation, and the development process.
- Increased knowledge and availability of early autism screening resources, benefiting families and medical professionals alike.
- Possible basis for later advancement into a comprehensive screening instrument or connection with medical databases.

- The project aims to assist parents and medical professionals in early identifying autism in children, enabling prompt support and intervention.

Project Keywords

Autism Spectrum Disorder (ASD)

Machine Learning

Early Detection

Predictive Modeling

Healthcare Technology

Chapter 03: Research Gap

Recent studies have utilized AI and machine learning to predict autism risk in early diagnosis. Using algorithms like SVM, Decision Trees, Random Forests, and Deep Learning, these studies have improved the precision and effectiveness of autism screening. However, there are still research gaps in this area, making the proposed project necessary.

- This paper (Farooq, 2023) develops a dataset-driven prediction model focused on ethical standards and data privacy while addressing parental concerns. It evaluates different machine learning techniques on a local dataset to propose a practical framework for diagnosing Autism Spectrum Disorder (ASD) in educational and medical settings.
- This (Ahmed, 2022) develops a model for early detection of Autism Spectrum Disorder (ASD) that is ethically sound, dataset-based, privacy compliant, and scalable, aimed at creating a useful prediction system for clinical and educational settings.
- This paper (Mukherjee, 2023) aims to develop an accurate and ethical autism spectrum disorder (ASD) prediction model using structured data and advanced machine learning techniques, addressing gaps in ethical data management noted in prior studies.
- Using a Random Forest model, the study (M.Abdelwahab, 2024) identified autism spectrum disorder (ASD) with 99.75% accuracy but neglected ethical considerations.

- In contrast, the current project aims to create a child-centered and ethical approach for detecting ASD, focusing on accessible and privacy-preserving early screening frameworks for infants.

Chapter 04: Requirement Analysis

Requirements analyses are crucial for software creation, ensuring the system meets user expectations and system goals. The "Early Detection of Autism in Children using Machine Learning" project relies on accurate data collection and evaluation.

The requirements for an autism screening system were determined through various techniques, including parent discussions, interviews, and research, and categorized into functional, non-functional, and user categories.

Functional Requirements

Functional requirements outline the system's functions, including user input gathering, machine learning data processing, and autism risk prediction.

Functional Requirement	Description
Data Input Module	Parents or physicians can input child data such as age, gender, behavioral observations, and developmental milestones into the system.
Data Validation	Before processing, the system must verify that all data entered is accurate and complete.
Machine Learning Prediction	Based on input features, the system will forecast the probability of autism using the trained machine learning model.
Result Generation	The system will generate prediction results and indicators, such as "Low Risk," "Moderate Risk," or "High Risk."
Report Generation	The system will generate a summary report of the prediction that can be viewed or downloaded for additional medical consultation.

Model Comparison Module	The system will enable researchers or developers to accurately assess machine learning algorithms like SVM, Random Forest, and Decision Tree.
Data Storage	Anonymized user data will be safely stored by the system for model retraining and performance enhancement.
Admin Dashboard	An administrative interface for managing users, datasets, and model updates will be part of the system.

Non-Functional Requirements

Non-functional requirements define the system's quality attributes, including security, usability, performance, and dependability, ensuring its effectiveness, security, and usability.

Non-Functional Requirement	Description
Performance	After data is submitted, the system should deliver prediction results in five seconds.
Accuracy	During validation testing, the prediction model's accuracy should remain at least 85%.
Usability	Users without technical expertise should be able to use the interface because it should be easy to understand and simple.
Security	All users and child data must adhere to data protection regulations and be stored safely.
Scalability	To support more users or datasets in the future, the system should be scalable.
Maintainability	Modularity in the system would make it simple to replace or update the machine learning models.
Portability	The system ought to function across various platforms, including mobile devices and web browsers.
Reliability	The system ought to guarantee reliable operation free from frequent crashes and data loss.

User Requirements

From the standpoint of the end users, who in this project are mainly parents, physicians, and system administrators, user requirements specify the expectations and interactions.

User Requirement	Description
Ease of use	Parents shouldn't require technical or medical knowledge to enter behavioral and developmental data.
Interpretability of results	Predictions ought to be presented by the system in an easily comprehensible manner (e.g., color-coded risk levels).
Guidance and recommendations	Following prediction, the system ought to offer broad suggestions or, in the event of high risk, recommend speaking with an expert.
Confidentiality	Only authorized users should have access to user data, including child details, which should be kept private.
Feedback mechanism	Users ought to be able to comment on the predictability or usability of the system.
Doctor access	Physicians ought to be able to examine reports for several patients and monitor development over time.
System updates	When a better model or updated version is released, the system ought to alert users.

Chapter 05: Finance

A budget plan is required for the project "Early Detection of Autism in Children using Machine Learning," including estimated costs for data processing, software, hardware, human resources, and other indirect costs.

The proposed system consists of a prototype application for parents and physicians, data preprocessing, and machine learning model development, requiring research, development, software tools, and computer resources.

Budget Estimation and Cost Categorization

Item	Description	Cost Category	Estimated Cost (LKR)
Software tools and Libraries	Python, ML libraries (TensorFlow, scikit-learn, Pandas), and IDEs (VS Code, PyCharm)	Direct	2000
Domain and Web Hosting	Prototype system hosting for a year (if you deploy online)	Direct	10000
Internet Usage	For obtaining research papers, datasets, and cloud access when required	Indirect	5000
User Testing/ Feedback sessions	Testing on a small scale with parents and physicians	Indirect	10000
Documentation/ Printing	Creating the final report's charts and diagrams	Indirect	3000
Maintenance/ Updates	After testing, the system underwent minor enhancements.	Indirect	5000

Financial Justification

The estimated budget of LKR 35,000–40,000 ensures cost-effective development of the early autism detection system using free datasets and personal computing resources. It covers essential software, minimal hosting, user testing, and documentation, supporting model training, prototype deployment, and evaluation while maintaining academic standards and practical feasibility.

Chapter 06: External Organizations

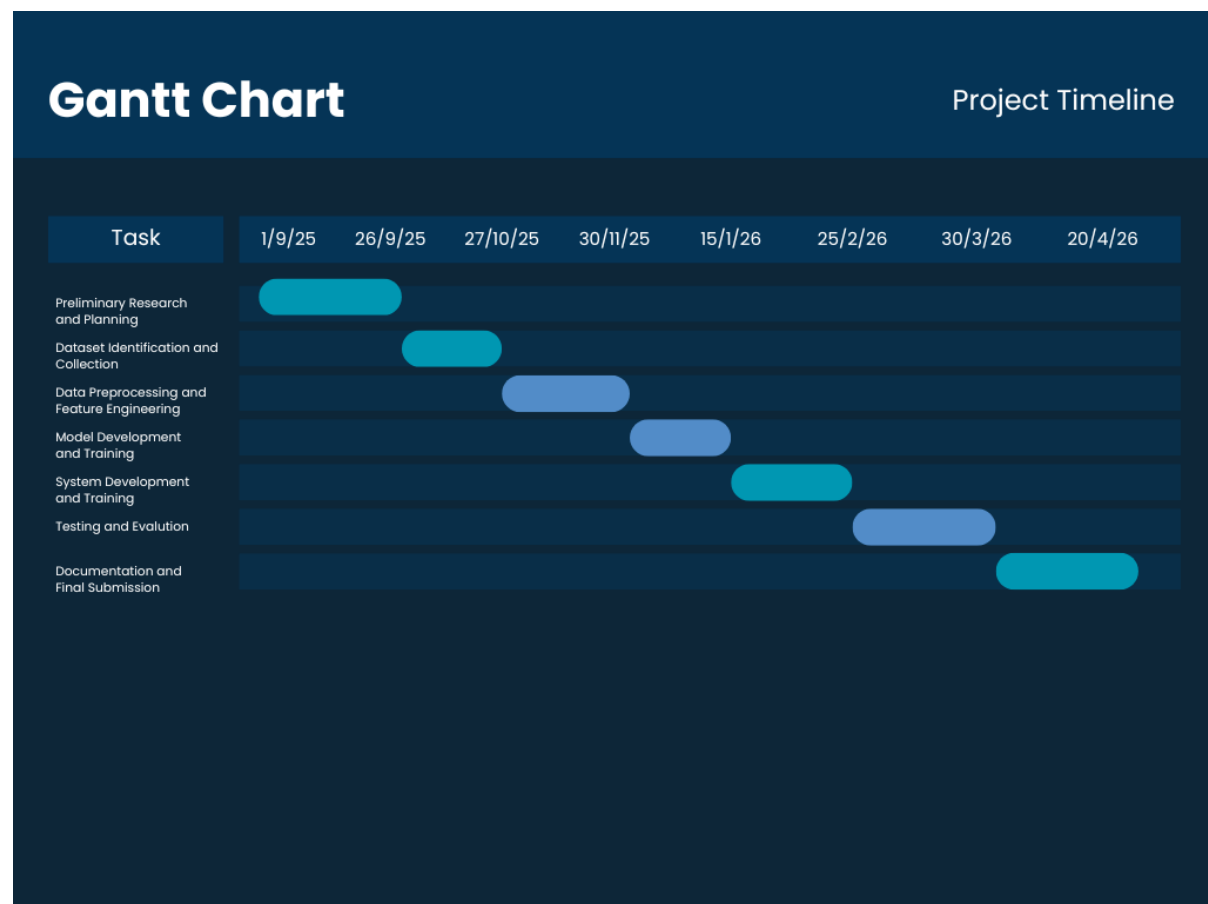
The "Early Detection of Autism in Children using Machine Learning" project relies on collaborations with experts in child psychology, autism research, healthcare, and technical development for societal significance and scientific reliability.

Dr. Lasantha Wijesekara (Medical doctor, Applied Psychologist)

Role - Offered experienced guidance on the psychological and behavioral characteristics of autism spectrum disorder (ASD). provided recommendations on identifying trustworthy behavioral indicators for the dataset, interpreting symptoms, and ethical process.

Contribution - Helped guarantee that the ML model complies with rules of ethics when managing sensitive data and is in line with real-life human behavior assessment standards.

Chapter 07: Time Frame/ Timeline



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